

Final Exam

ECON 320 – Introduction to Mathematical Economics

December 10, 2025

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Instructions:

- You are allowed **120 minutes** to complete the exam.
 - Show all work. Unsupported answers receive little credit.
 - Unless stated otherwise, all variables are real and differentiable as needed.
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Multiple Choice Questions (20 points)

Each question is worth 2 points.

1. Suppose demand is $Q = \sqrt{P + 10} - I$. The derivative $\partial Q / \partial I$ equals:
 - (a) $\frac{1}{2}(P + 10)^{-1/2}$
 - (b) -1
 - (c) 0
 - (d) $-\frac{1}{2}(P + 10)^{-1/2}$
2. For $z = x^2y^3$, the total differential dz is:
 - (a) $2xy^3dx + 3x^2y^2dy$
 - (b) $2xydx + 3x^2ydy$
 - (c) $2xy^3 + 3x^2y^2$
 - (d) $dx + dy$
3. Let $f(x) = 4x - x^3$. At $x = \sqrt{4/3}$, the function has:
 - (a) A local maximum
 - (b) A local minimum
 - (c) An inflection point
 - (d) Cannot be determined
4. For $Y = AK^{0.5}L^{0.5}$, the elasticity of Y with respect to K is:
 - (a) 0.5
 - (b) 1
 - (c) $\frac{L}{K}$
 - (d) 0

5. For $f(x, y) = 3x^2 + 4xy + y^2$, the Hessian implies:
 - (a) Saddle point
 - (b) Local minimum
 - (c) Local maximum
 - (d) Cannot classify

6. In $\max U(x, y) = xy$ subject to $x + y = 10$, the Lagrange multiplier represents:
 - (a) Marginal utility of x
 - (b) Marginal utility of y
 - (c) Marginal utility of relaxing the constraint
 - (d) Share of income allocated to x

7. In $\max_x f(x, a) = ax - x^2$, the envelope theorem gives:
 - (a) x
 - (b) ax
 - (c) $2ax$
 - (d) x^2

8. Consumer surplus for demand $P(Q) = 20 - Q$ at price $P = 10$ is:
 - (a) 25
 - (b) 50
 - (c) 100
 - (d) 150

9. The solution to $\frac{dy}{dt} = 4y$ with $y(0) = 3$ is:
 - (a) $y = 3 + 4t$
 - (b) $y = 3e^{4t}$
 - (c) $y = 12t$
 - (d) $y = 4e^{3t}$

10. If $Y(t) = K^{0.3}L^{0.7}e^{0.02t}$, then:
 - (a) $\frac{dY}{dt} = 0.02Y$
 - (b) $\frac{dY}{dt} = 0.3Y$
 - (c) $\frac{dY}{dt} = 0.7Y$
 - (d) $\frac{dY}{dt} = 0$

Problems (80 points total)

Problem 1 (16 points)

A firm produces output

$$Q = x^{1/2}y^{1/3}.$$

1. Compute the total differential dQ . (5 pts)
2. If the firm controls only x , compute dQ/dx . (3 pts)
3. Profit is $\pi(x) = 40Q - x - y - 5$. Holding y fixed, find the optimal x . (5 pts)
4. Determine whether the optimum is a maximum or minimum. (3 pts)

Problem 2 (16 points)

A technology firm's revenue evolves as

$$R(t) = 500e^{0.03t}(1 + t^{1/2}).$$

1. Compute $\frac{dR}{dt}$. (5 pts)
2. Compute the growth rate $g(t) = \frac{1}{R} \frac{dR}{dt}$. (4 pts)
3. Compute the elasticity of revenue with respect to time. (4 pts)
4. Determine whether revenue growth accelerates or decelerates over time. (3 pts)

Problem 3 (16 points)

Consider

$$f(x, y) = 20x - 4x^2 + 10y - 3y^2 - xy.$$

1. Compute the first-order partial derivatives. (4 pts)
2. Solve the first-order conditions. (4 pts)
3. Compute the Hessian matrix and its determinant at the critical point. (4 pts)
4. Classify the critical point. (4 pts)

Problem 4 (16 points)

A consumer maximizes

$$U(x, y) = x^{1/2}y^{1/2}$$

subject to

$$p_x x + p_y y = I.$$

1. Set up the Lagrangian and derive the first-order conditions. (5 pts)
2. Solve for x^* and y^* . (5 pts)
3. Use the envelope theorem to compute $\frac{\partial V}{\partial I}$. (3 pts)
4. Interpret the result. (3 pts)

Problem 5 (16 points)

Capital accumulation satisfies

$$\frac{dK}{dt} = I(t) - \delta K,$$

where

$$I(t) = 50e^{0.01t}, \quad \delta = 0.05.$$

1. Solve the differential equation for $K(t)$. (6 pts)
2. Find the particular solution when $K(0) = 100$. (4 pts)
3. Compute $\int_0^{20} I(t) dt$. (3 pts)
4. Interpret the long-run behavior of $K(t)$. (3 pts)